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Technical White Paper: Distributed SD-WAN Overlay Multicast Validation with Stigix

Validating IGMPv2/v3, PIM-SM/SSM, Rendezvous Point (RP) Convergence, and Real-Time RFC 3550 Telemetry Across Prisma SD-WAN Fabrics

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Executive Summary

Enterprise multicast applications—such as live IP video distribution, financial market data feeds, push-to-talk (PTT) radio systems, and industrial IoT telemetry—pose unique operational challenges over modern Software-Defined Wide Area Networks (SD-WAN). Unlike unicast flows, multicast relies on a multi-layered state machine spanning host kernel IGMP memberships, LAN Layer-3 transit routing (e.g., VyOS / core switches), First-Hop Router (FHR) PIM registration, Rendezvous Point (RP) arbitration, SD-WAN fabric tunnel replication, and Last-Hop Router (LHR) delivery.

A single missing link in this chain—such as an unconfigured intermediary PIM hop, a mismatched IGMP version, or a TTL threshold—results in silent packet blackholing that standard ping or TCP synthetic probes cannot detect.

This white paper details the architecture, control/data plane separation, network engineering configuration (Prisma SD-WAN ION & VyOS), and end-to-end telemetry methodology of the **Stigix SD-WAN Overlay Multicast Validation Engine**.

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1. Architectural Foundations: Control Plane

vs. Data Plane Separation

A core design principle of Stigix is the strict decoupling between the **Orchestration Control Plane** (Stigix Fleet Mesh) and the **Network Validation Data Plane** (Native Kernel UDP/IGMP).

```
flowchart TB
    subgraph ControlPlane["1. STIGIX CONTROL PLANE (Encrypted Web
        UI["Operator Web Browser\n(SSE Real-Time Stream)"]
        Leader["DC1-Ubuntu (LEADER)\nOrchestrator & Telemetry Agg
        PeerBR5["BR5-Ubuntu (PEER)\nBranch Agent\n192.168.217.5"]
        PeerBR8["BR8-Ubuntu (PEER)\nBranch Agent\n192.168.219.1"]
        UI <==>|"HTTPS / SSE"| Leader
        Leader <==>|"Inbound/Outbound WS\nControl & Telemetry"| P
        Leader <==>|"Inbound/Outbound WS\nControl & Telemetry"| P
    end

    subgraph DataPlane["2. PHYSICAL & SD-WAN MULTICAST DATA PLANE
        SenderNIC["DC1 Host Kernel (ens9)\n192.168.203.100\nRaw U
        VyOS["Intermediary Transit Router (VyOS)\neth5 (192.168.2
        DC1_ION["Prisma SD-WAN DC1-R1 ION\nPort 1: 192.168.201.3\
        Fabric["Prisma SD-WAN Secure Fabric\n(BR5-INET Overlay Tu
        BR5_ION["Prisma SD-WAN BR5 ION\nSub-Interface 217 on Port
        ReceiverNIC["BR5 Host Kernel (enp2s0)\n192.168.217.5\nIGM

        SenderNIC -->|"L2 MAC 01:00:5e:01:01:64"| VyOS
        VyOS -->|"PIM-SM (S,G) Forwarding"| DC1_ION
        DC1_ION ===>|"SD-WAN Overlay"| Fabric
        Fabric ===>|"Replicated Stream"| BR5_ION
        BR5_ION <-->|"IGMPv3 Report / Multicast UDP"| ReceiverNIC
    end
```

1.1 How Leader-to-Peer Coordination Works Under the Hood

1. **Zero-Touch Enrollment & Profile Synchronization**:
    - Branch nodes (such as `BR5-Ubuntu` and `BR8-Ubuntu`) join the `DC1-Ubuntu` Leader using a cryptographic **Magic Join Token** (`STX-...`).

- Each Peer establishes a persistent WebSocket tunnel (`INBOUND WS`) to the Leader (`192.168.203.100:8080`).
- Whenever a Multicast Profile is created or modified on the Leader, the configuration bundle is pushed across the Fleet Mesh so all nodes share a unified topology awareness.

2. **Coordinated Stream Execution (`Start Stream`)**:

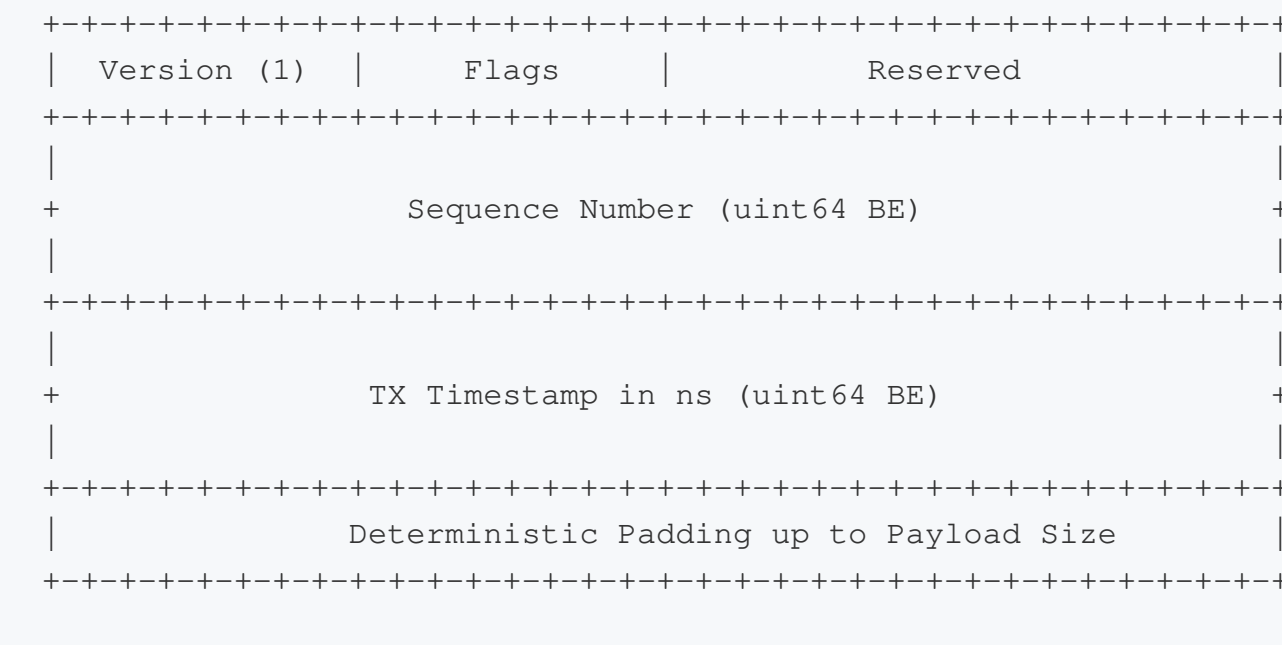
- **Step A — Branch IGMP Subscription (Receiver Readiness)**: The Leader sends an RPC command over the WebSocket control channel to all participating Receiver nodes (`BR5-Ubuntu`). The receiver opens a native UDP socket bound to `0.0.0.0:5001` and executes a kernel `setsockopt(IP_ADD_MEMBERSHIP)` (ASM) or `MCAST_JOIN_SOURCE_GROUP` (SSM) on its local LAN interface (`enp2s0`, `192.168.217.5`). The Linux kernel immediately transmits an **IGMPv2/v3 Membership Report** onto the branch LAN.
- **Step B — Paced Sender Injection**: After a `250 ms` tree-priming delay, the Sender node (`DC1-Ubuntu`) begins transmitting Constant Bitrate (CBR) UDP packets out of its physical interface (`ens9`, `192.168.203.100`) to `239.1.1.100:5001` with `IP_MULTICAST_TTL = 8` and `DSCP = EF (0xb8)`.
- **Step C — Real-Time Telemetry Backhaul**: Multicast payload packets **never** traverse the Stigix WebSocket control tunnel. They must cross the real physical routers and SD-WAN overlay. Every second, the Receiver node computes its local reception metrics and streams a lightweight JSON telemetry frame back to the Leader over the WebSocket control channel.

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2. Binary Wire Protocol & Telemetry Mathematics

Every UDP multicast packet transmitted by Stigix carries a deterministic **24-byte `STGX` binary header** at the start of the UDP payload:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                          Magic ("STGX")                       |
```



2.1 4096-Packet Sliding Bitmask Window (Loss, Duplicates, Reordering)

Rather than relying on naive packet counters that fail during out-of-order delivery or late joins:

- **Late-Join Baseline (`firstSeq`)**: When a receiver joins mid-stream (e.g., at sequence `7087`), it anchors its expected packet counter at `firstSeq = 7087`, preventing false-positive loss reports for packets sent prior to the IGMP Join.
- **Bitmask Tracking**: A 4096-slot circular bitmask tracks every sequence number:
    - If a sequence number is already marked `1` in the bitmask, Stigix increments the **Duplicate (`Dup`)** counter (often caused by dual-tunnel SD-WAN replication anomalies or L2 loops).
    - If a packet arrives with `seq < maxSeq`, Stigix increments the **Out-of-Order (`Reord`)** counter and marks the bitmask slot, dynamically healing the loss count.

2.2 RFC 3550 Inter-Arrival Jitter

Stigix calculates real-time network jitter using the standard RTP/RTCP **RFC 3550** exponentially weighted moving average formula, which is immune to clock skew between the Sender and Receiver hosts:

$$D(i, j) = (R_j - R_i) - (S_j - S_i)$$

$$J_j = J_{j-1} + \frac{|D(i, j)| - J_{j-1}}{16}$$

Where S_i, S_j are the sender nanosecond timestamps from the `STGX` header and R_i, R_j are the receiver monotonic arrival timestamps.

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3. Spoke-to-Spoke Topology: How a BR8 (Source) $\rightarrow$ BR5 (Receiver) Test Works

Because Stigix separates orchestration from data generation, any Peer node in the fleet can act as the **Multicast Source (Sender)** while any subset of Peers acts as **Receivers**—without routing multicast traffic through the Stigix Leader (`DC1`).

```
sequenceDiagram
    autonumber
    participantDC1 as Stigix Leader (DC1-Ubuntu)
    participantBR8 as Sender Peer (BR8-Ubuntu)
    participantION8 as Prisma ION BR8 (FHR)
    participantRP as Prisma ION DC1-R1 (RP 192.168.201.3)
    participantION5 as Prisma ION BR5 (LHR)
    participantBR5 as Receiver Peer (BR5-Ubuntu)

    Note over participantDC1,participantBR5: Phase 1 — Stigix Mes
    participantDC1->>participantBR5: WS RPC: Join Multicast Group
    participantDC1->>participantBR8: WS RPC: Start UDP Stream to

    Note over participantBR8,participantBR5: Phase 2 — Shared Tre
    participantBR5->>participantION5: Kernel IGMPv3 Membership Re
    participantION5->>participantRP: SD-WAN Overlay PIM (*,G) Joi
    participantBR8->>participantION8: Raw UDP Multicast (S=192.16
    participantION8->>participantRP: Unicast PIM Register (Encaps
    participantRP->>participantION5: Forward down (*,G) Shared Tr
    participantION5->>participantBR5: Deliver Multicast UDP to er

    Note over participantION8,participantION5: Phase 3 — SPT Swit
    participantION5->>participantION8: PIM (S,G) Join directly ov
    participantION8->>participantION5: Direct Branch-to-Branch Mu
    participantBR5-->>participantDC1: WS Telemetry: RX Bitrate, I
    participantBR8-->>participantDC1: WS Telemetry: TX Bitrate, S
```

1. **Orchestration Role of `DC1`**: When the operator selects `Sender: BR8-Ubuntu` and `Receiver: BR5-Ubuntu` on the `DC1` dashboard, `DC1` sends start instructions to `BR8` and `BR5` via WebSocket and aggregates their live telemetry reports.
2. **Initial RPT Forwarding (`*,G`)**: `BR5` ION sends a PIM `(*,G)` Join to the Static RP (`192.168.201.3` on `DC1-R1`). Meanwhile, `BR8` ION receives the multicast stream on its LAN and sends unicast **PIM Register** packets to `DC1-R1`.
3. **SPT Switchover (Direct Spoke-to-Spoke)**: Because **SPT Switchover** is enabled on the Prisma SD-WAN fabric, as soon as `BR5` ION receives the first packet and learns the source IP of `BR8` (`192.168.219.1`), it initiates a source-specific `(S,G)` tree directly toward `BR8` ION, optimizing latency by bypassing the hub once the tree converges.

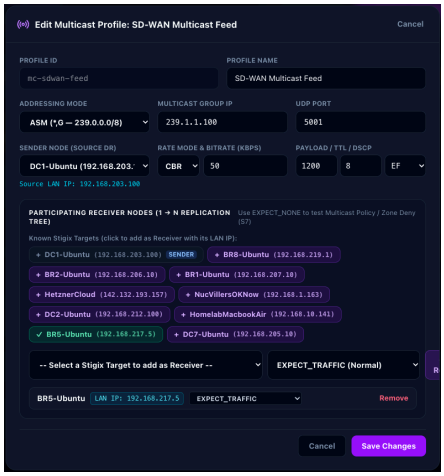
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4. End-to-End Reference Lab Configuration (`DC1` $\rightarrow$ `BR5`)

Below is the complete, production-validated configuration across all four layers of the lab: **Stigix**, **Prisma SD-WAN Branch ION (`BR5`)**, **Prisma SD-WAN Hub ION (`DC1-R1`)**, and the **Intermediary Transit Router (`VyOS`)**.

4.1 Layer 1 — Stigix Multicast Profile Configuration

In the Stigix Web Dashboard (**Custom Apps $\rightarrow$ Multicast Overlay**), the profile is configured with the following parameters:



Parameter	Configured Value	Engineering Rationale

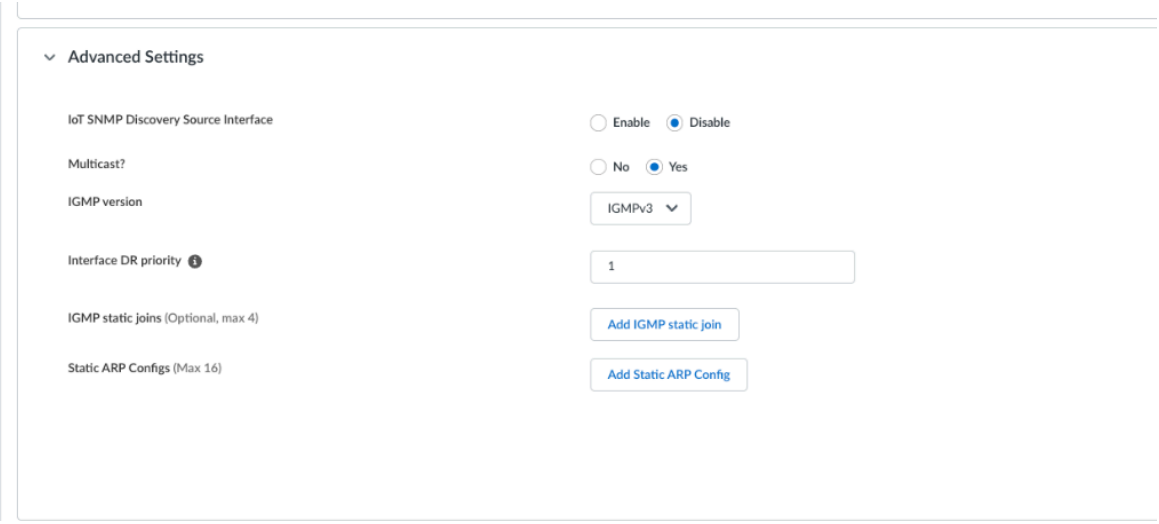
Profile ID / Name	`mc-sdwan-feed` / `SD-WAN Multicast Feed`	Identifies the synthetic multicast test profile across all cluster nodes.
Addressing Mode	`ASM (*,G — 239.0.0.0/8)`	Tests Any-Source Multicast using the Prisma SD-WAN Rendezvous Point (`192.168.201.3`).
Multicast Group & Port	`239.1.1.100 : 5001`	Administratively scoped organization-local multicast group.
Sender Node (Source DR)	`DC1-Ubuntu (192.168.203.100)`	Transmits on physical interface `ens9`.
Rate Mode & Bitrate	`CBR 50 kbps` (`1200 B` payload, `5 pps`)	Constant predictable pacing (`200 ms` inter-packet gap) ideal for sequence tracking.
TTL & DSCP	`TTL = 8`, `DSCP = EF`	`TTL 8` survives multiple L3 hops (`DC1 -> VyOS -> DC1-R1 -> BR5-ION -> BR5`); `EF` maps to real-time voice/video QoS.
Receiver Node	`BR5-Ubuntu (192.168.217.5)`	Configured with policy `EXPECT_TRAFFIC`.

4.2 Layer 2 — Prisma SD-WAN Branch ION Configuration (`BR5`)

On the branch router (`BR5`), three multicast components must be active:

A. LAN Sub-Interface IGMPv3 Activation

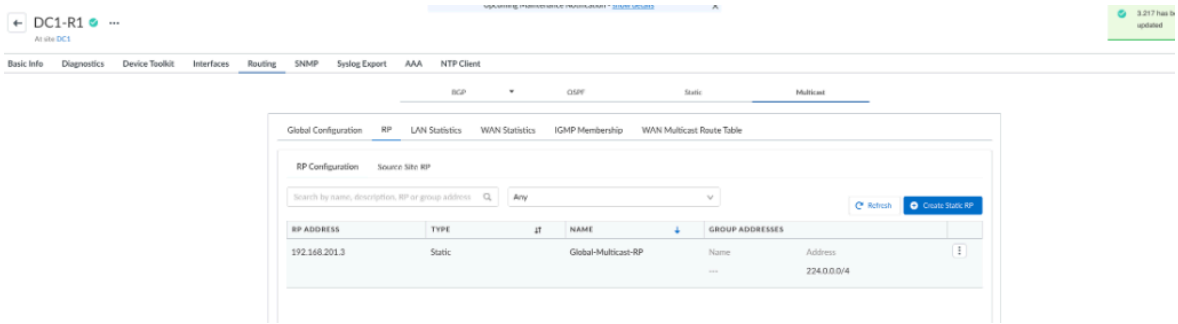
Under **Interfaces $\rightarrow$ Port 3 $\rightarrow$ Sub-Interface (217) (`192.168.217.254/24`) $\rightarrow$ Advanced Settings**, enable **Multicast (`Yes`)** and set **IGMP version** to **`IGMPv3`**:



> [!TIP] Linux kernels (`Ubuntu 22.04/24.04`) default to sending **IGMPv3 Membership Reports** (`proto 2`, type `0x22` to `224.0.0.22`) when `IP_ADD_MEMBERSHIP` is invoked. Setting the Prisma SD-WAN LAN interface to **IGMPv3** ensures immediate membership processing without requiring kernel `force_igmp_version` overrides on Linux hosts.

B. Static Rendezvous Point (RP) Definition

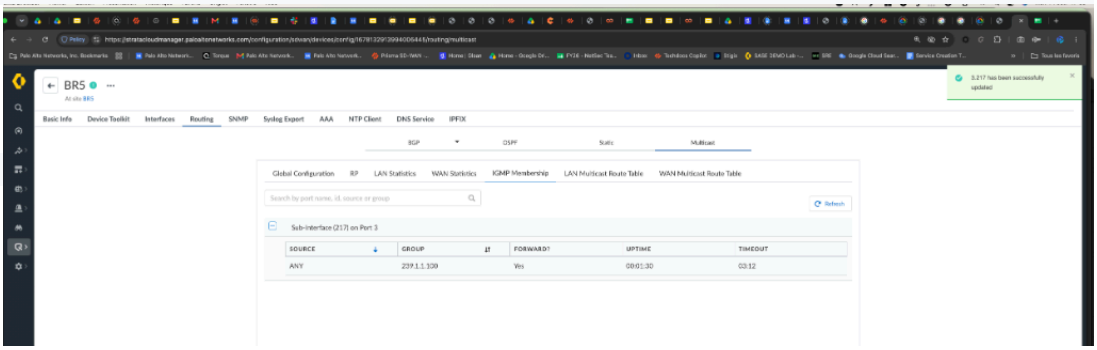
Under **Routing $\rightarrow$ Multicast $\rightarrow$ RP**, define the global Rendezvous Point pointing to `DC1-R1`'s LAN IP (`192.168.201.3`) for `224.0.0.0/4`:



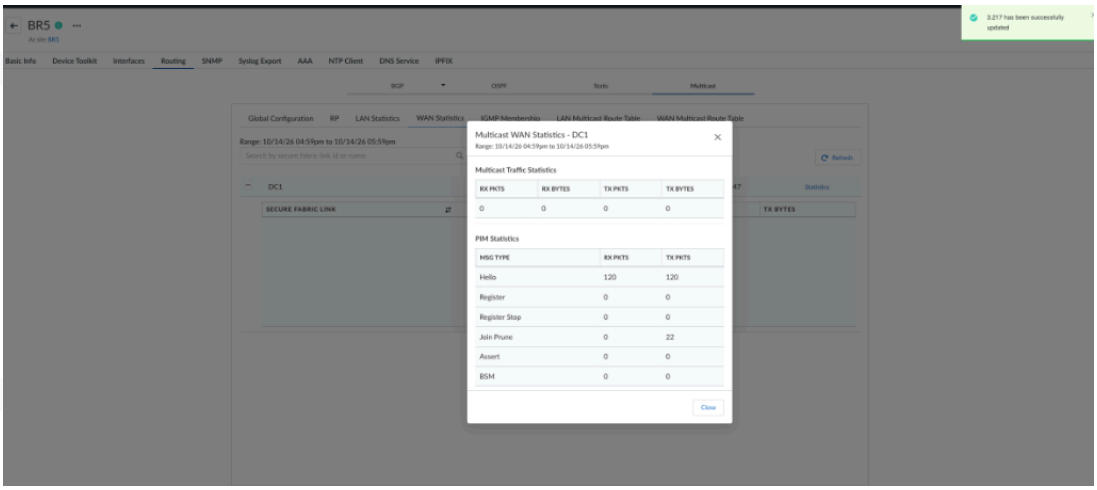
C. Verifying Branch Control Plane State (`BR5`)

Once Stigix starts the stream, `BR5` ION confirms local IGMP reception and upstream WAN PIM signaling:

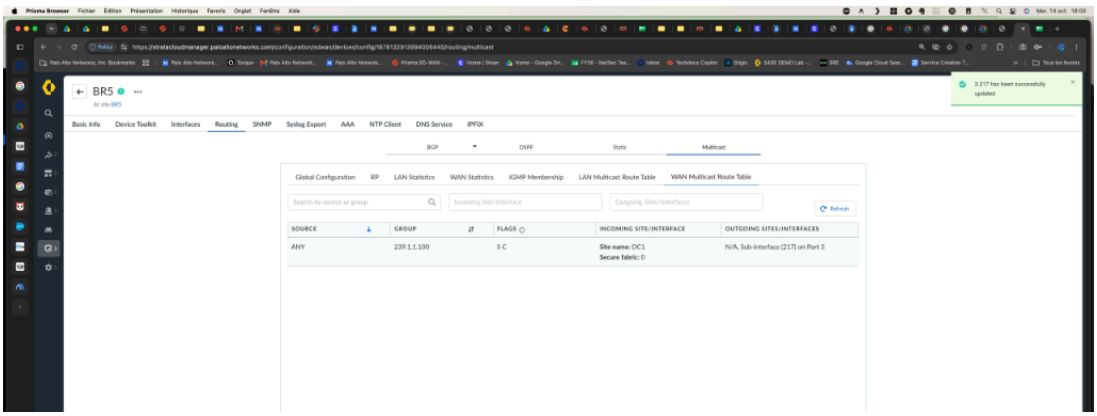
1. **IGMP Membership Table** (`Sub-Interface (217) on Port 3` shows `GROUP: 239.1.1.100`, `SOURCE: ANY`, `FORWARD?: Yes`):



2. **WAN Multicast Statistics toward `DC1`** (`Join Prune TX PKTS: 22` sent over the SD-WAN fabric to `DC1`):



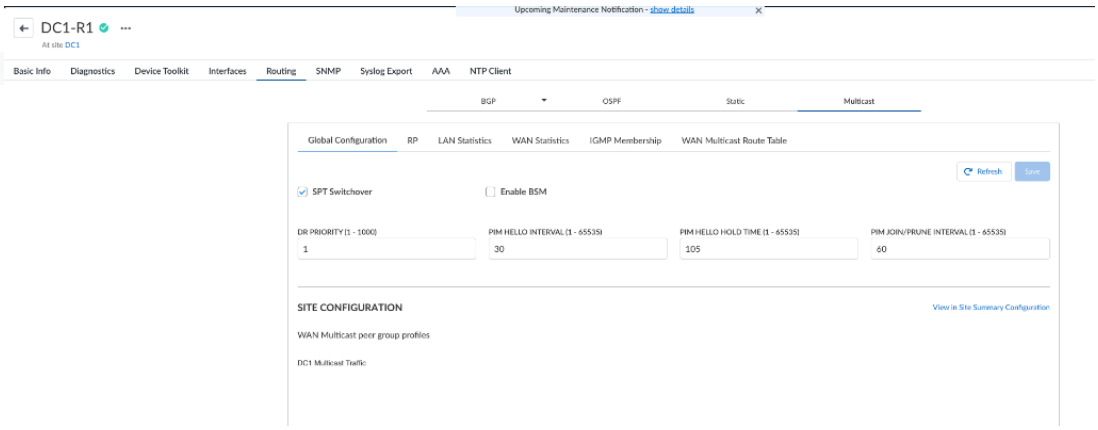
3. **WAN Multicast Route Table on `BR5`** (`FLAGS: S C`, `INCOMING: Site name: DC1`, `OUTGOING: Sub-Interface (217) on Port 3`):



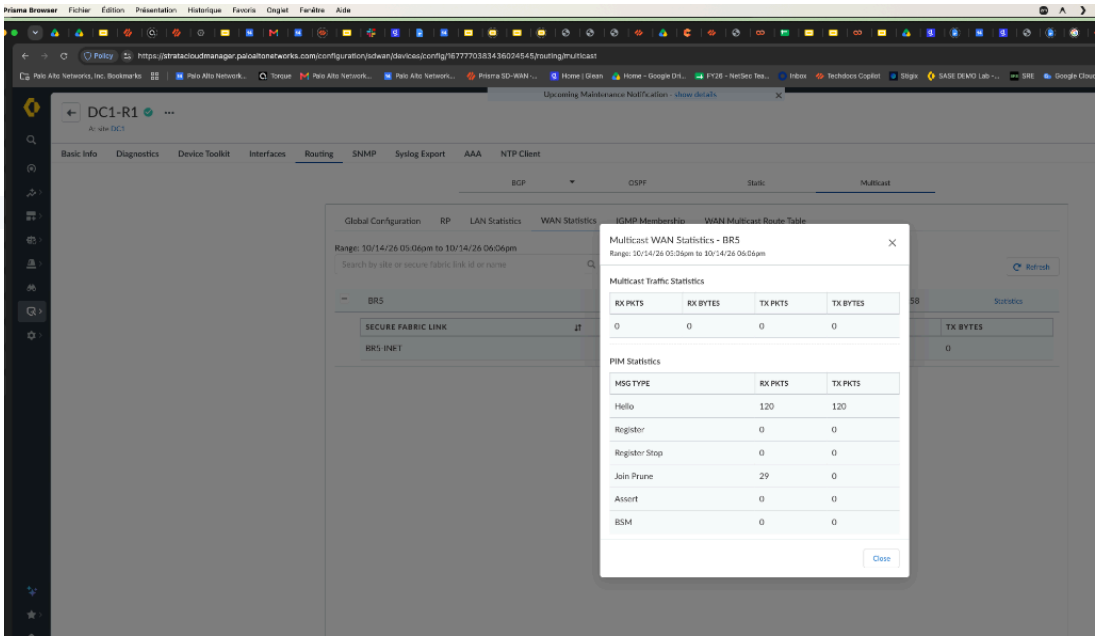
4.3 Layer 3 — Prisma SD-WAN Hub ION Configuration (`DC1-R1`)

At the Data Center hub (`DC1-R1`), the ION acts as both the **Static Rendezvous Point (`192.168.201.3`)** and the **Hub Multicast Replicator**:

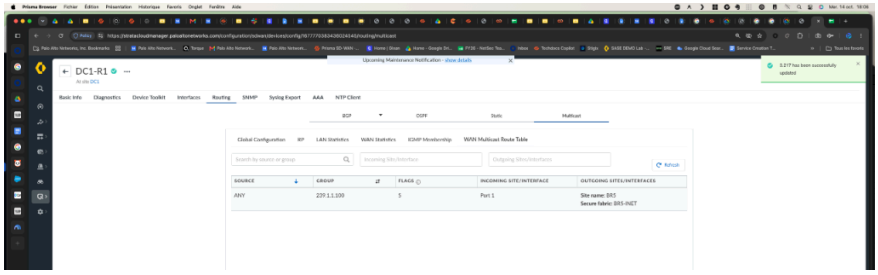
1. **Global Multicast Configuration**: Enable **SPT Switchover** and bind the **WAN Multicast peer group profile** (`DC1 Multicast Traffic`):



2. **WAN PIM Join Reception from `BR5`**: Under **WAN Statistics (`BR5-INET`)**, `DC1-R1` confirms receiving `Join Prune RX PKTS: 29` from `BR5`:



3. **WAN Multicast Route Table on `DC1-R1`**: Installs the active replication state (`GROUP: 239.1.1.100`, `INCOMING: Port 1`, `OUTGOING: Site name: BR5, Secure fabric: BR5-INET`):



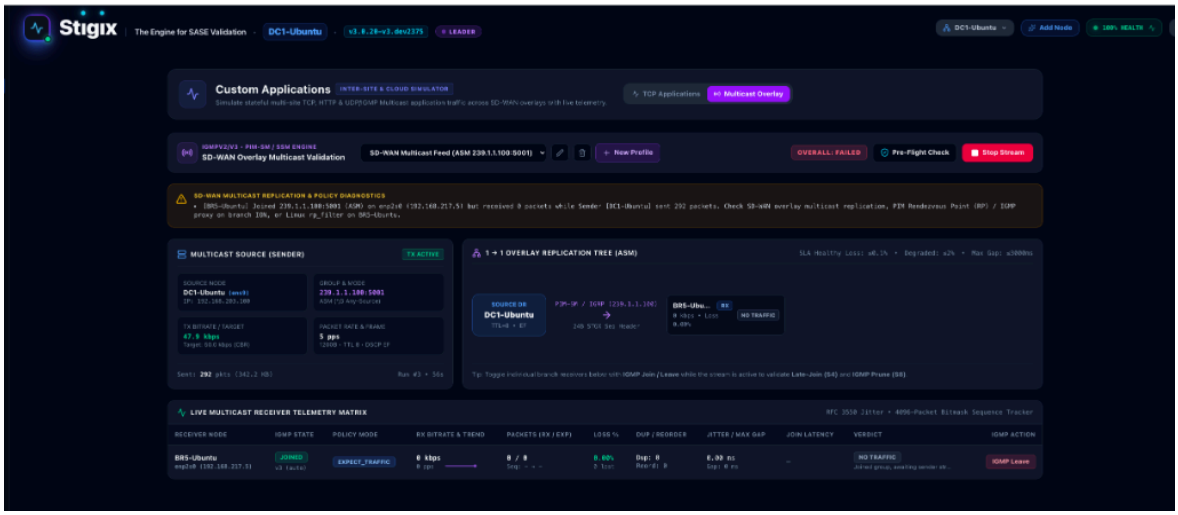
4.4 Layer 4 — The Critical Intermediary Transit Router (VyOS)

In many enterprise Data Centers, application servers (`DC1-Ubuntu` on `192.168.203.100/24`) do not sit on the same Layer-2 broadcast domain as the SD-WAN ION LAN interface (`DC1-R1 Port 1` on `192.168.201.3/24`). An intermediary core router (`vyoslandc1`) routes traffic between them:

- `eth5` (`192.168.203.10/24`): Default gateway for `DC1-Ubuntu` (`192.168.203.100`).
- `eth1` (`192.168.201.10/24`): Transit LAN connected to `DC1-R1` (`192.168.201.3`).

Symptom When PIM Is Missing on the Intermediary Router

When `vyoslandc1` only runs BGP unicast without PIM-SM enabled on `eth5` and `eth1`, `VyOS` silently drops all multicast packets arriving on `eth5`. Stigix immediately flags this condition with an automated **Overlay Blackhole Diagnostic**:



Enabling PIM-SM on VyOS (`vyoslandc1`)

To bridge the multicast tree between the `DC1-Ubuntu` Sender subnet (`eth5`) and the Prisma SD-WAN RP (`eth1`), apply the following configuration on VyOS:

```
configure
set protocols pim interface eth5
set protocols pim interface eth1
set protocols pim rp address 192.168.201.3 group '224.0.0.0/4'
commit
```

```
save
exit
```

Verifying Multicast State on VyOS

Immediately after committing, VyOS establishes a PIM adjacency with `DC1-R1` (`192.168.201.3`) on `eth1` and installs the `(192.168.203.100, 239.1.1.100) (S,G)` entry with flags `SFT` (Sparse, Register, SPT-bit set):

```
vyos@vyoslandc1:~$ show ip pim neighbor
 Interface  Neighbor       Uptime    Holdtime  DR Pri
 eth1       192.168.201.3  00:00:42  00:01:33  1

vyos@vyoslandc1:~$ show ip pim interface
 Interface  State  Address         PIM Nbrs  PIM DR  FHR  IfChanr
 eth1       up     192.168.201.10  1         local   0    1
 eth5       up     192.168.203.10  0         local   0    0
 pimreg     up     0.0.0.0         0         local   0    0

vyos@vyoslandc1:~$ show ip multicast route
IP Multicast Routing Table
Flags: S - Sparse, C - Connected, P - Pruned
       R - SGRpt Pruned, F - Register flag, T - SPT-bit set
 Source           Group        Flags  Proto  Input  Output  TTL
 192.168.203.100  239.1.1.100  SFT    PIM    eth5   eth1    1
```

---

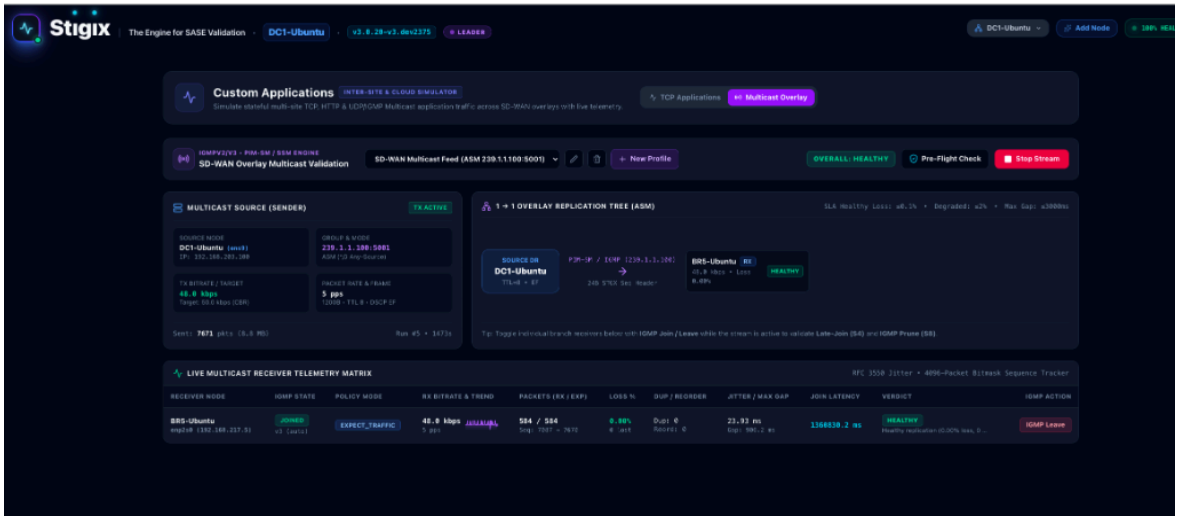
5. Live Telemetry Results & SLA Diagnostics in Stigix

Once the full multicast path is operational, Stigix provides continuous millisecond-level visibility into the health of the SD-WAN replication tree:

5.1 Healthy Replication State (`OVERALL: HEALTHY`)

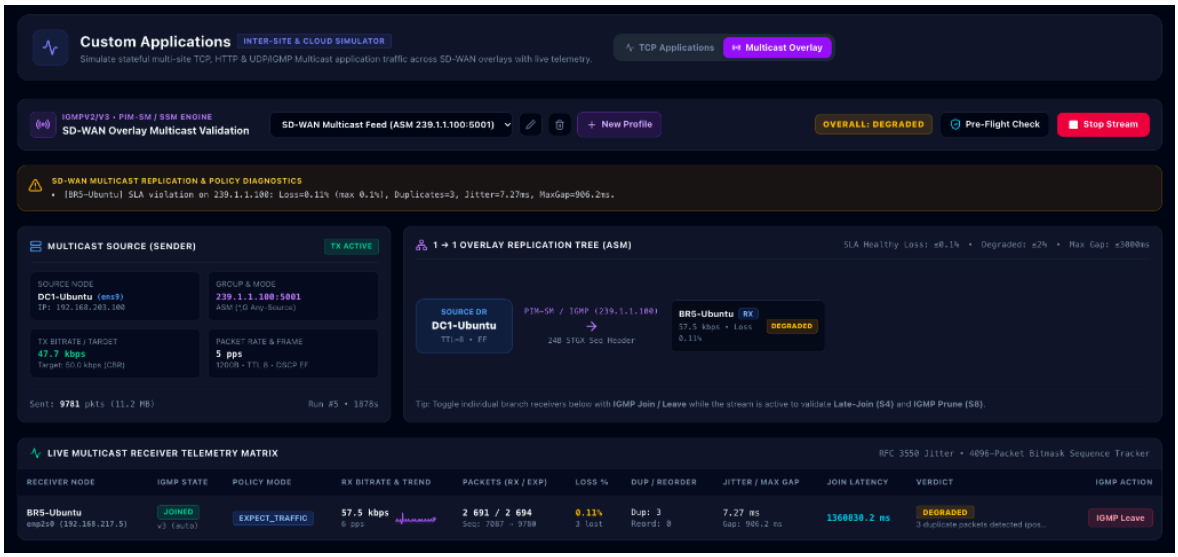
In steady state, `BR5-Ubuntu` receives `584 / 584` packets (`0.00%` loss, `0` duplicates, `0` reordered) at the exact target rate (`48.0 kbps`, `5 pps`) with `23.93 ms` RFC 3550

jitter:



5.2 Real-Time SLA Violation Detection (`OVERALL: DEGRADED`)

When transient WAN impairments or tunnel switchovers occur across the SD-WAN fabric, Stigix's 4096-bit sequence tracker catches single-packet anomalies that traditional SNMP counters miss. In the example below, over a `2,694`-packet run, Stigix detected `3 lost packets (0.11% loss)` and `3 duplicate packets (Dup: 3)`, automatically transitioning the verdict to `DEGRADED` with a specific SLA diagnostic alert:



🏷 Revision History

Date	Stigix Version	Author / Trigger	Summary of Changes
2026-10-14	`v3.0.21`	Stigix Core Team	Initial publication of the SD-WAN Overlay Multicast Validation Technical White Paper with Prisma SD-WAN ION, VyOS PIM-SM, and Stigix telemetry architecture